DIGITAL LOGIC DESIGN — COURSE PROJECT

EEG-Based Sleep Stage Classifier

Combinational & Sequential Logic for Biomedical Signal Classification

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Course: Digital Logic Design (DLD)

Submitted to: Ms. Meezan Hamid

Overview

The **EEG-Based Sleep Stage Classifier** is a digital logic design project that identifies human sleep stages — Deep Sleep, Light Sleep, Relaxed State, and Awake State — based on simulated EEG frequency band activity. Rather than processing real analog EEG signals, the system uses DIP switches to represent the activation of the four primary EEG bands (Delta, Theta, Alpha, Beta), and combinational and sequential digital logic to classify and display the corresponding sleep stage.

The design integrates a priority encoder, sequential state storage, a stable clock source, and a seven-segment display driver into a single working biomedical-inspired digital system, verified first in simulation and then on physical hardware.

Objectives

- Design a digital logic circuit for EEG sleep stage classification
- Simulate EEG frequency bands using DIP switch inputs
- Implement priority logic using the 74LS148 priority encoder
- Develop a sequential digital system using D flip-flops
- Generate synchronized clock pulses using an NE555 timer
- Display classified sleep stages on a seven-segment display
- Verify system operation through both Proteus simulation and hardware implementation

Background

Electroencephalography (EEG) records the electrical activity generated by neurons in the brain, and is widely used in sleep analysis, neurological diagnosis, seizure monitoring, and brain–computer interfaces. Different sleep and wakefulness states are characterized by distinct dominant EEG frequency bands:

EEG Band	Frequency Range	Physiological State
Delta (δ)	0.5 – 4 Hz	Deep Sleep
Theta (θ)	4 – 8 Hz	Light Sleep
Alpha (α)	8 – 13 Hz	Relaxed / Drowsy
Beta (β)	13 – 30 Hz	Awake / Alert

This project models that classification process in hardware: when multiple EEG bands are active simultaneously, the system resolves the conflict using priority logic that reflects real physiological dominance — higher-frequency activity overrides lower-frequency activity.

System Architecture

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DIP Switches (EEG Bands) → 74LS148 Priority Encoder → 74LS04 Inverter
→ 74LS74 D Flip-Flops (State Storage) → 74LS47 Decoder → 7-Segment Display
↑
NE555 Astable Timer (Clock)
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1. DIP switches simulate EEG band activation, stabilized with pull-up resistors.
2. The 74LS148 priority encoder determines the highest-priority active EEG band.
3. The 74LS04 inverter converts the encoder's active-low outputs to active-high.
4. D flip-flops store the current sleep stage between clock pulses.
5. The NE555 timer, in astable mode, generates the synchronizing clock (~1 Hz).
6. The 74LS47 decoder drives a common-anode seven-segment display showing the classified stage.

Methodology

Priority Encoding — The 74LS148 resolves conflicts when multiple EEG bands are active at once, using the priority order **Beta > Alpha > Theta > Delta**, matching real physiological dominance of higher-frequency brain activity.

Sequential Logic — D flip-flops give the system memory, transforming it from a purely combinational circuit into a sequential one capable of holding its classified state between clock edges.

Clock Generation — An NE555 timer in astable configuration provides a stable ~1 Hz clock to synchronize state updates.

Display Logic — The 74LS47 BCD-to-seven-segment decoder converts the stored binary state into a visible digit representing the current sleep stage.

Priority Truth Table

B	A	T	D	S1	S0	Sleep Stage
1	X	X	X	1	1	Awake
0	1	X	X	1	0	Relaxed
0	0	1	X	0	1	Light Sleep
0	0	0	1	0	0	Deep Sleep

Boolean expressions: $S1 = B + A$ $S0 = B + T$

State Mapping

S1	S0	Sleep Stage
0	0	Deep Sleep
0	1	Light Sleep
1	0	Relaxed
1	1	Awake

Components Used

Component	Part Number	Qty	Function
Priority Encoder	74LS148	1	Encodes dominant EEG band
Hex Inverter	74LS04	1	Converts active-low outputs
D Flip-Flop	74LS74	1	Stores sleep stage state
BCD to 7-Segment Decoder	74LS47	1	Drives display
Timer IC	NE555	1	Generates clock signal
DIP Switch	DIPSW-4	1	Simulates EEG bands
7-Segment Display	Common Anode	1	Displays sleep stage
Resistors	10 kΩ, 68 kΩ	Multiple	Pull-up and timing resistors
Capacitors	0.01 μF, 10 μF	2	Timing capacitors
Breadboard	—	1	Hardware assembly
Connecting Wires	—	Multiple	Circuit connections

Results

The system was first designed and verified in **Proteus**, then implemented on a **breadboard** for hardware testing. Both stages confirmed correct classification across all sleep states.

Active Input	Dominant Band	Display Output	Sleep Stage
1000	Beta	3	Awake
0100	Alpha	2	Relaxed
0010	Theta	1	Light Sleep
0001	Delta	0	Deep Sleep
1100	Beta	3	Awake
0110	Alpha	2	Relaxed
0000	None	0	Deep Sleep

Parameter	Result
Priority Logic	Successful
State Storage	Successful
Display Output	Correct
Clock Synchronization	Stable
Hardware Verification	Successful
Proteus Simulation	Successful

The priority encoder correctly selected the dominant EEG band in every test case, the NE555 timer produced stable clock pulses, the D flip-flops held state reliably between transitions, and the seven-segment display accurately represented all four classified sleep stages with no significant glitches observed.

Discussion

Real EEG systems must resolve multiple simultaneously active frequency bands — the priority encoder models this by consistently selecting the physiologically dominant band. Adding D flip-flops moved the design from purely combinational logic to a true sequential system capable of retaining state, which is essential for any system that must track a condition over time rather than react only to instantaneous inputs. While the classifier uses simulated digital inputs rather than real analog EEG signals, it captures the core logic used in biomedical signal classification and offers a low-cost, hands-on way to learn combinational logic, sequential logic, state machines, timing circuits, and biomedical signal interpretation together.

Relevance to Biomedical Engineering

- **EEG Signal Classification** — mirrors the simplified logic behind neurological monitoring systems used in hospitals and research labs
- **Sleep Disorder Monitoring** — sleep stage detection underlies diagnosis of conditions such as sleep apnea, narcolepsy, insomnia, and REM sleep disorders
- **Biomedical Instrumentation** — integrates sensing, encoding, memory storage, timing, and display, mirroring the subsystems of real biomedical devices
- **Embedded Biomedical Systems** — introduces the digital logic foundations behind modern wearable EEG devices

Advantages and Limitations

Advantages

- Low-cost implementation
- Easy to understand and demonstrate
- Real-time operation
- Combines biomedical and digital logic concepts
- Expandable architecture

Limitations

- Uses simulated EEG inputs instead of real analog signals
- Limited number of sleep stages
- Simplified classification approach with no spectral analysis
- Single-channel representation only

Future Enhancements

- Integrating real EEG sensors and analog signal conditioning circuits
- ADC-based digitization of live EEG signals
- FPGA-based architecture
- Machine learning-based classification
- Alarm and monitoring system integration
- Wireless data transmission
- Multi-channel EEG analysis
- Real-time patient monitoring dashboard

Conclusion

The EEG-Based Sleep Stage Classifier successfully demonstrated the application of digital logic design principles to a biomedical engineering problem. The 74LS148 priority encoder correctly resolved conflicts between multiple active EEG bands according to defined priority rules, while the D flip-flops and NE555 timer introduced stable sequential behavior and state storage. Both Proteus simulation and breadboard implementation verified correct system operation across all four sleep stages, demonstrating a cost-effective way to combine biomedical concepts with digital electronics and providing a foundation for more advanced biomedical instrumentation projects.

Repository Structure

```
EEG-Sleep-Stage-Classifier/
├── README.md
├── Report/
│   └── DLD_Project_Report.pdf
├── Circuit/
│   ├── Priority_Encoder/
│   ├── Flipflop_Logic/
│   └── Display_Driver/
├── Simulation/
│   └── Proteus_Files/
├── Images/
│   └── Project_Photos/
└── Results/
    └── Truth_Tables/
```

Add your Proteus simulation files, circuit diagrams, breadboard photographs, and test results to the relevant folders above.

Author

Name	Roll No
Muhammad Salman Tahir	2024-BME-39

Submitted to: Ms. Meezan Hamid

References

1. Iber, C. et al. (2007). *The AASM Manual for the Scoring of Sleep and Associated Events*. American Academy of Sleep Medicine.
2. Texas Instruments (2004). *74LS148 8-Line to 3-Line Priority Encoder Datasheet*.
3. Texas Instruments (2004). *74LS74 Dual D-Type Positive-Edge-Triggered Flip-Flop Datasheet*.
4. Texas Instruments (1988). *NE555 Precision Timer Datasheet*.
5. Texas Instruments (2003). *74LS47 BCD to Seven-Segment Decoder Datasheet*.
6. Proteus Design Suite — Labcenter Electronics.

7. Floyd, T. L. *Digital Fundamentals*. Pearson Education.
8. Webster, J. G. *Medical Instrumentation: Application and Design*. Wiley.